

Uri Cohen

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INTRODUCTION

ABOUT ME

I am an aspiring scientist in the field of computational neuroscience, interested in developing theories of the dynamics of learning and memory in the brain and putting them to the test through collaboration with experimentalists. With a strong mathematical background and hands-on experience with advanced statistical physics, computer science, and statistics methods, I approach problems and experimental results in neuroscience. Currently focusing on biological memory.

PHD THESIS

Title: *Analysis of invariant object representations through linear classification of manifolds.*

Advisor: Professor Haim Sompolsky

In my PhD thesis, I used statistical physics methods to analyze neuronal responses to objects and measure the properties important for linear object classification. Those are geometric properties of object manifolds which I use to shed light on object representations in artificial deep networks and the brain, and how they change across levels of processing.

RESEARCH INTERESTS

- Population coding and manifold representation of sensory information; the transformation of manifold representations through biological learning.
- Memory encoding with biological learning, the dynamics of memory retrieval, and memory consolidation; their relation to hippocampal phenomenology (theta modulation, sharp wave ripples, behavioral timescale synaptic plasticity, sleep stages, etc.).
- The interplay and functional role of reward-modulated plasticity, homeostasis plasticity, Hebbian, and anti-Hebbian plasticity. Biological alternatives to error backpropagation, or biological implementation of gradient descent.
- Neural implementation of world-models, namely inference of state changes in the environment, and their possible contribution to predictive sensory and control schemes.

EDUCATION AND RESEARCH

2026 Postdoctoral fellow at Weizmann Institute of Science

with Professor Yaniv Ziv at the Department of Brain Sciences.

2023 – 2025 Postdoctoral visiting fellow at the University of Cambridge

with Professor Máté Lengyel at the Computational and Biological Learning Lab.

2022 Postdoctoral fellow at the Hebrew University of Jerusalem

with Professor Haim Sompolsky at the Edmond & Lily Safra Center for Brain Sciences.

2011 – 2021 MSc and PhD at the Hebrew University of Jerusalem

PhD program in Brain Sciences at the Edmond & Lily Safra Center for Brain Sciences.

2002 – 2005 BSc at the Hebrew University of Jerusalem

Graduated Summa Cum Laude a double major in Mathematics and Cognitive Sciences.

AWARDS

- 2026 Recipient of the **Faculty Dean Fellowship** at Weizmann Institute of Science.
- 2022 Recipient of the **Blavatnik Cambridge Postdoctoral Fellowship**.
- 2011 Recipient of the Hebrew University **Rector's award** (equivalent to 1st in class).
- 2005 Graduated **Summa Cum Laude** (average grade 97.2) a BSc at the Hebrew University.
- 2004-2005 **Amirim Nature**, an excellence program for Hebrew University students.
- 2004 Recipient of the Hebrew University **Rector's award** (equivalent to 1st in class).
- 2003 Recipient of the Hebrew University **Rector's award** (equivalent to 1st in class).
- 1995 **Gur Arye prize** for high-school graduation projects for my work on *Statistical analysis and game theory of the mastermind game*.

ACTIVITIES

- **Co-organizer of a satellite workshop** for the Bernstein Conference 2026 on *Plasticity and function of hippocampal memory circuits* (Frankfurt am Main, September 2026)
- **Co-organizer** of the *Deep Learning Journal Club* at the Jerusalem Deep Learning Center.
- **Conference talks:**
 - Physics for Deep Learning (Tel Aviv, Israel, June 2024, invited speaker)
 - Cambridge Memory Meeting 2024 (Cambridge, UK, May 2024)
 - International Workshop on Learning and Information Theory 2023 (Shenzhen & Hong Kong, China, July 2023, invited speaker)
 - Gatsby meeting 2022 (Hebrew University, Israel, May 2022)
- **Conference presentations (posters):**
 - *Theory of dynamical stability in attractor neural networks predicts CA3 neurons' activation functions*. Bernstein Conference 2026.
 - *A theory of memory stability in hippocampal area CA3*. COSYNE 2024.
 - *Geometric properties of face representations in the macaque face patch system and deep neural networks*. COSYNE 2019.
 - *Computational benefits of sensory expectation cancellation*. ISFN 2014.
- **Reviewer** for the *Computational and Systems Neuroscience* (COSYNE) conference.
- **Reviewer** for top machine learning conferences: *Conference on Neural Information Processing Systems* (NeurIPS) 2022-6 ([top reviewer in 2024](#)), *International Conference on Machine Learning* (ICML) 2023-5, *International Conference on Learning Representations* (ICLR) 2022-6, *Association for the Advancement of Artificial Intelligence* (AAAI) 2026-7.
- **Reviewer** for the *NeurIPS workshop on Symmetry and Geometry in Neural Representations*.
- **Reviewer** for American Physical Society's *Physical Review Letters* (PRL), *Physical Review X life* (PRX life), *Physical Review X intelligence* (PRX intelligence), and *Physical Review E* (PRE).
- **Reviewer** for *Journal of Machine Learning Research* (JMLR), *Transactions on Machine Learning Research* (TMLR), *IEEE Transactions on Neural Networks and Learning Systems*
- **Curator** of the [Random Matrix Gallery](#) website.
- **Teaching experience:** served for seven years as a teaching assistant for the following courses:
 - *Statistical physics of learning and memory* (introductory seminar for physics students): mentoring students in preparing their talks and giving several lectures.
 - *Statistical mechanics of spin glasses and neural networks*: giving several lectures.
 - *Topics in the theory of deep learning*: a weekly lecture and preparing exercises.
- Extensive experience with **neural data analysis**, including: [Freiwald & Tsao 2010](#), [Alme, et al. 2014](#), [Majaj, Hong, et al. 2015](#), [Cohen, et al. 2017](#), [Stringer, Pachitariu, et al. 2019](#), [Reber, et al. 2019](#), [Malezieux, et al. 2020](#), [Sheintuch, et al. 2023](#), [Li, et al. 2024](#), [Chettih, et al. 2025](#).

PUBLICATIONS

- **Cohen, U.***, Picher, M.*, Navas Olivé, A., & Lengyel, M. (2026). *The activation function of CA3 pyramidal neurons is optimal for the stable recall of memory patterns*. bioRxiv preprint.
- **Cohen, U.**, & Lengyel, M. (2025). *Dynamical stability for dense patterns in attractor neural networks*. arXiv preprint (under review for PNAS).
- **Cohen, U.** (2025). *Eigenvalue spectrum support of paired random matrices with pseudo-inverse*. arXiv preprint.
- **Cohen, U.**, & Sompolinsky, H. (2022). *Soft-margin classification of object manifolds*. Physical Review E, 106(2), 024126.
- Froudarakis, E., **Cohen, U.**, Diamantaki, M., Walker, E. Y., Reimer, J., Berens, P., Sompolinsky, H., & Tolias, A. S. (2020). *Object manifold geometry across the mouse cortical visual hierarchy*. bioRxiv (reviewed manuscript in eLife, 2026).
- **Cohen, U.***, Chung, S. *, Lee, D. D., & Sompolinsky, H. (2020). *Separability and geometry of object manifolds in deep neural networks*. Nature Communications, 11(1), 1-13.
- Chung, S., **Cohen, U.**, Sompolinsky, H., & Lee, D. D. (2018). *Learning data manifolds with a cutting plane method*. Neural Computation, 30(10), 2593-2615.
- Mattioni, M., **Cohen, U.**, & Le Novère, N. (2012). *Neuronvisio: a graphical user interface with 3d capabilities for NEURON*. Frontiers in Neuroinformatics, 6, 20.

REFERENCES

- **Professor Haim Sompolinsky**: Center for Brain Science, *Harvard*; Edmond & Lily Safra Center for Brain Sciences, *The Hebrew University of Jerusalem*. haim@fiz.huji.ac.il
- **Professor Daniel D. Lee**: Tisch University Professor of Electrical and Computer Engineering, *Cornell Tech*; Executive Vice President, *Samsung Research* ddl46@cornell.edu
- **Professor Máté Lengyel**: Computational and Biological Learning Lab, Department of Engineering, *University of Cambridge*. m.lengyel@eng.cam.ac.uk

INDUSTRY EXPERIENCE

- **2007 – 2010 Senior Team Lead and Senior Software Developer at Microsoft R&D**
Leading a team of 4 software developers and handling all professional aspects: mentoring and guiding professional growth, project management, and technical leadership.
- **2005 – 2007 Senior Software Developer at Kidaro (an Israeli start-up)**
The leading technical person from project inception and building the company's work processes, through design and implementation of a complete product, to being acquired by Microsoft.

SKILLS

- Hold a broad perspective on the field of computational neuroscience.
- Thorough, down-to-details approach.
- Passion for teaching, extensive experience.
- Welcomes feedback and criticism.
- Superb software engineering skills.
- Master advanced methods of statistical physics, optimization theory, and Bayesian modeling.
- Experience with deep learning tools, numerical methods, and large-scale simulations.
- Believe in experimental testing of theory and collaboration with experimentalists.

* Equal contribution co-first authors